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Mathematical Modelling of the Three-Dimensional Free Vibration of Bidirectionally Graded Porous Plates on a Kerr Foundation: A Rayleigh–Ritz Eigenvalue Approach

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Abstract: This paper develops a mathematical model for the three-dimensional (3D) free vibration of porous plates whose material properties are graded along the length and through the thickness and which rest on a three-parameter Kerr foundation. The problem is formulated as an eigenvalue problem for the 3D elasticity operator with coefficients that vary in two coordinates, and it is solved by the Rayleigh–Ritz method. The trial functions are numerically orthonormalised Legendre polynomials, the separable structure of the coefficients turns every matrix block into a Kronecker product, and the massless foundation field is eliminated by static condensation, which leads to a symmetric generalized matrix eigenvalue problem. The model is verified against an independent exact solution, obtained for thickness-dependent coefficients from a two-point boundary-value problem for six first-order differential equations, and against classical results for homogeneous plates; its convergence is examined for different edge conditions. Two reduced two-dimensional models, a higher-order shear deformation theory and a reduced-unknown quasi-3D theory, are formulated in the same framework and assessed against the 3D model. The results show that in-plane grading lowers the eigenvalues considerably and makes the eigenfunctions asymmetric, that a stiff foundation can turn an in-plane mode into the fundamental mode, and that porosity lowers the frequency of an unsupported plate but raises it on a stiff foundation. The higher-order theory agrees closely with the 3D model for simply supported plates of moderate and small thickness, whereas the quasi-3D theory retains a noticeable error even for very thin thickness-graded plates because of thickness locking. New 3D benchmark eigenvalues are provided.

Keywords: Mathematical modelling; Rayleigh–Ritz method; Generalized eigenvalue problem; State-space method; Functionally graded plates; Free vibrations; Kerr foundation

1 Introduction

Many problems of applied mechanics lead to eigenvalue problems for partial differential operators with variable coefficients. The free vibration of a plate whose material properties change from point to point is a typical example: the natural frequencies and mode shapes are the eigenvalues and eigenfunctions of the three-dimensional (3D) elasticity operator, whose coefficients follow the spatial distribution of the stiffness and the density. Closed-form solutions exist only when the coefficients depend on a single coordinate and the boundary conditions allow a separation of variables. In all other cases the problem must be approximated, and two questions then arise: how to construct a discretisation that is accurate, well conditioned and verifiable, and how much accuracy is lost when the 3D problem is replaced by a simpler two-dimensional (2D) model.

Functionally graded materials (FGMs) give rise to exactly such problems. Their composition changes continuously, typically from a ceramic on one face to a metal on the other [1]. When the service conditions also vary along a component, the composition can be graded in a second direction, which gives more freedom to tailor the stiffness and the thermal stresses than one-directional grading [2, 3]. The static and dynamic behaviour of these bidirectional FGMs has been studied for beams [4, 5] and plates [6–11]. Mathematically, grading along the length as well as through the thickness makes the coefficients of the governing equations depend on two coordinates. The Fourier-series solutions available for thickness-graded plates then no longer apply, even for simply supported edges.

For plates graded through the thickness only, the free-vibration problem is well understood. Vel and Batra [12] obtained exact 3D solutions for simply supported plates, Uymaz and Aydogdu [13] used a 3D Ritz method for other edge conditions, and Matsunaga [14] and Hosseini-Hashemi et al. [15] gave higher-order and first-order plate solutions. The Rayleigh–Ritz method with polynomial trial functions is particularly suited to such problems [13, 16, 17]: it handles arbitrary combinations of edge conditions, it converges from above to the exact eigenvalues, and the trial functions need to satisfy only the essential boundary conditions. Its main numerical difficulty is the ill-conditioning of the mass and stiffness matrices when high-degree polynomials are used, which limits the attainable accuracy. Reduced 2D models are an alternative. Shear deformation theories and quasi-3D theories, which add a transverse normal strain, have been proposed with full expansions [18, 19] and with a reduced number of unknowns [20–24]. They replace the 3D eigenvalue problem by a smaller one, but their accuracy has to be established against 3D solutions. Plates on two- and three-parameter elastic foundations have been analysed with such models [25–27], and Zhou et al. [16] gave 3D Ritz solutions for thick homogeneous plates on a Pasternak foundation.

For bidirectionally graded plates the available results are based on reduced models only. Melaibari et al. [7] studied the free vibration of bidirectional graded porous plates on elastic foundations with a 2D differential quadrature method, and Lieu et al. [11] analysed the bending and free vibration of in-plane bidirectional plates by isogeometric analysis. Kerr-foundation studies have addressed the thermal vibration of bidirectionally stepped porous plates [28] and the free vibration of thickness-graded porous plates with a quasi-3D theory [29]. To the author's knowledge, no 3D solutions are available for the free vibration of plates graded along the length and through the thickness on a Kerr foundation. Without them the accuracy of the reduced models cannot be assessed. This matters because, for the same plates under mechanical load, the reduced-unknown quasi-3D theory was found to be slightly stiffer than the 3D solution because of thickness locking, and in-plane grading shifted the peak deflection away from the plate centre [30]. Both effects can be expected to influence the eigenvalues and eigenfunctions as well.

This paper develops and verifies a mathematical model for this eigenvalue problem. The main contributions are as follows:

1. A 3D Rayleigh–Ritz formulation is constructed for a plate whose stiffness and density vary in two coordinates and which rests on a three-parameter Kerr foundation. The trial functions are Legendre polynomials with boundary factors, orthonormalised numerically to keep the matrices well conditioned. The separable structure of the coefficients is exploited so that every matrix block becomes a Kronecker product. The massless foundation field is eliminated by static condensation, which leads to a symmetric generalized eigenvalue problem for the plate alone.
2. An independent exact solution is derived for the case in which the coefficients depend on the thickness coordinate only. The problem is reduced to a two-point boundary-value problem for a system of six first-order ordinary differential equations, and the eigenvalues are obtained as roots of a boundary determinant. This solution, together with classical results for homogeneous plates, verifies the Ritz model, and its convergence is examined for different edge conditions.
3. Two reduced 2D models, a higher-order shear deformation theory (HSDT) and a reduced-unknown quasi-3D theory, are formulated within the same Rayleigh–Ritz framework and assessed against the 3D model over a range of thickness ratios, gradation indices and edge conditions.
4. New 3D benchmark eigenvalues are provided, and a parametric study shows how the gradation indices, porosity, the three foundation parameters, the edge conditions and the thickness ratio affect the eigenvalues and the shape of the eigenfunctions. A kinetic-energy criterion is introduced to distinguish flexural from in-plane modes.

The paper is organised as follows. Section 2 presents the mathematical model, Section 3 the Rayleigh–Ritz discretisation, Section 4 the verification, the assessment of the reduced models and the parametric study, and Section 5 the conclusions.

2 Mathematical Model

Consider a rectangular plate of length a, width b and thickness h, occupying $0 \le x \le a$, $0 \le y \le b$, $-h/2 \le z \le h/2$, which rests on a three-parameter Kerr foundation (Figure 1). The plate and the foundation follow the model developed in a previous study [30], and only the ingredients needed for free vibration are summarised here.

2.1 Bidirectional Porous Material

The ceramic volume fraction changes along the length and through the thickness,

$$V_c(x,z) = \left(\frac{x}{a}\right)^{k_x} \left(\frac{1}{2} + \frac{z}{h}\right)^{k_z}, \tag{1}$$

so that the bottom face and the edge $x = 0$ are pure metal, while the top face becomes pure ceramic at $x = a$. With an even porosity of volume fraction α_p, the modified rule of mixtures [31] gives the Young modulus and the mass density as

$$E = E_m + (E_c - E_m)V_c - \frac{\alpha_p}{2}(E_c + E_m),$$
$$\rho = \rho_m + (\rho_c - \rho_m)V_c - \frac{\alpha_p}{2}(\rho_c + \rho_m). \tag{2}$$

Poisson's ratio is constant, $\nu = 0.3$. The density follows the same rule as the modulus, as is customary for porous FG plates [7, 31]. Porosity therefore lowers both the stiffness and the inertia of the plate, but in different proportions: at the metal-rich bottom face it removes $\alpha_p(E_c + E_m)/(2E_m) = 3.2\alpha_p$ of the modulus and only $\alpha_p(\rho_c + \rho_m)/(2\rho_m) = 1.2\alpha_p$ of the density.

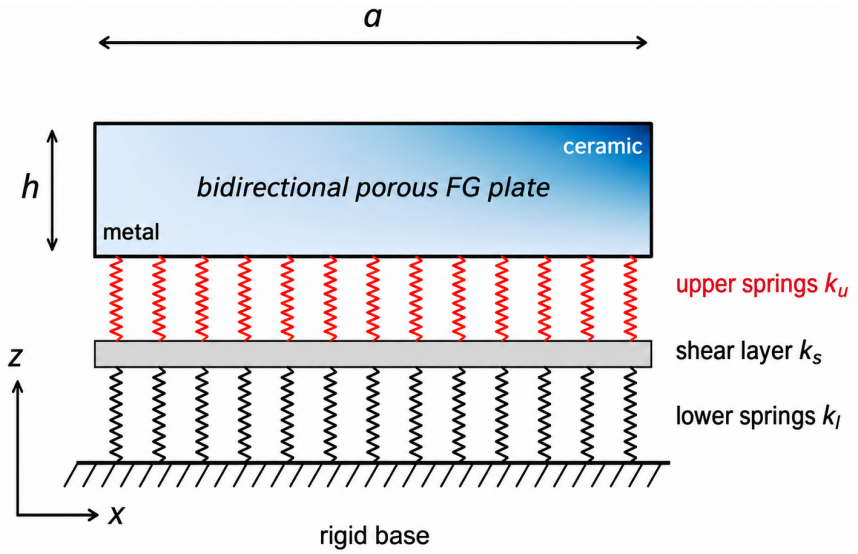


Figure 1. Bidirectional porous functionally graded (FG) plate on a three-parameter Kerr foundation (side view)
Note: FG = functionally graded; a = plate length; h = plate thickness; k_u = stiffness of the upper spring layer; k_s = stiffness of the shear layer; k_l = stiffness of the lower spring layer; the shading indicates the ceramic volume fraction.

2.2 Kerr Foundation

The Kerr foundation [32] comprises an upper spring layer of stiffness k_u, a shear layer of stiffness k_s and a lower spring layer of stiffness k_l. The deflection of the shear layer, $w_f(x, y)$, is kept as an independent field, and the foundation stores the energy:

$$U_f = \frac{1}{2}\int_0^a \int_0^b \left[k_u(w_0 - w_f)^2 + k_l w_f^2 + k_s(w_{f,x}^2 + w_{f,y}^2)\right] dx\, dy, \tag{3}$$

where, $w_0 = w(x, y, -h/2)$ is the deflection of the bottom face. The foundation is massless, so w_f carries no inertia. The Pasternak model [33] is recovered as $k_u \to \infty$, and the Winkler model [34] when, in addition, $k_s = 0$. The foundation parameters are written in the dimensionless form $K_u = k_u a^4/D_c$, $K_l = k_l a^4/D_c$, and $K_s = k_s a^2/D_c$, with $D_c = E_c h^3/[12(1 - \nu^2)]$.

2.3 Energy Functional

For harmonic motion with circular frequency ω, the displacements are $\{u, v, w\}e^{i\omega t}$ and $w_f e^{i\omega t}$. The free vibrations make the Rayleigh–Ritz functional

$$\Pi = \frac{1}{2}\int_V \varepsilon^T C \varepsilon \, dV + U_f - \frac{\omega^2}{2}\int_V \rho(u^2 + v^2 + w^2)\, dV \tag{4}$$

stationary, where $C(x, z) = E(x, z)\hat{C}(\nu)$ is the isotropic stiffness at each point. Eq. (4) contains no assumption about the through-thickness kinematics; the three models below differ only in how u, v and w are approximated.

2.4 Three-Dimensional Elasticity Model

In the 3D model each displacement component is expanded in all three coordinates,

$$\{u, v, w\} = \sum_{i,j,k} \{a_{ijk}, b_{ijk}, c_{ijk}\} X_i(\xi) Y_j(\eta) P_k(2\zeta - 1), \tag{5}$$

with $\xi = x/a$, $\eta = y/b$, $\zeta = z/h + 1/2$, and P_k the Legendre polynomial of degree k. The full 3D stiffness is used and the transverse normal and shear strains are retained, so the model is exact apart from the truncation of the series. Three-dimensional Ritz solutions with polynomial bases are well established for the vibration of thick homogeneous and FG plates [13, 16, 17]. The foundation acts on $w(x, y, -h/2)$. A simply supported edge is "hard", i.e. the tangential and transverse displacements vanish over the whole edge face.

2.5 Plate Models

The HSDT of Zenkour [35] and the reduced-unknown quasi-3D sinusoidal theory of Thai and Kim [20] are adopted in this study. The displacement field is

$$\begin{aligned} u &= u_0 - z w_{b,x} - f(z) w_{s,x}, \\ v &= v_0 - z w_{b,y} - f(z) w_{s,y}, \\ w &= w_b + w_s + g(z) w_z, \end{aligned} \tag{6}$$

with $f(z) = z - (h/\pi)\sin(\pi z/h)$ and $g(z) = 1 - f'(z) = \cos(\pi z/h)$. In the HSDT $w_z = 0$ and plane stress is assumed; the quasi-3D theory keeps w_z, the transverse normal strain $\varepsilon_z = g'(z) w_z$ and the full 3D stiffness. The kinetic energy is evaluated from Eq. (6) with all inertia terms, including the rotary and higher-order ones. Because E and ρ vary with x, the stiffness and inertia resultants are functions of x and are integrated numerically.

3 Rayleigh–Ritz Discretisation

The in-plane functions of all fields, including the shear-layer deflection w_f, are

$$X_i(\xi) = \xi^{p_0} (1 - \xi)^{p_1} P_i(2\xi - 1), \quad i = 0, \ldots, N - 1, \tag{7}$$

and similarly $Y_j(\eta)$. The exponents p_0 and p_1 impose only the essential conditions at each edge [30]. On an edge normal to x, $v = w = 0$ for a simply supported (S) edge and $u = v = w = 0$ for a clamped (C) edge (with, in the plate models, $w_{b,\,x} = w_{s,\,x} = 0$); a free (F) edge has no constraint. The shear-layer deflection vanishes on S and C edges. Edges are named in the order $x = 0$, $y = 0$, $x = a$, $y = b$. To keep the algebraic system well conditioned for large N, each set of one-dimensional (1D) functions is orthonormalised numerically before use; this changes the basis but not the space it spans.

Substituting the expansions into Eq. (4) and making Π stationary gives

$$\begin{bmatrix} K_{pp} & K_{pf} \\ K_{fp} & K_{ff} \end{bmatrix} \begin{bmatrix} c \\ e \end{bmatrix} = \omega^2 \begin{bmatrix} M & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c \\ e \end{bmatrix}, \tag{8}$$

where, c collects the plate coefficients and e those of w_f. Since the foundation is massless, e is condensed statically, which leaves the symmetric eigenvalue problem

$$(K_{pp} - K_{pf} K_{ff}^{-1} K_{fp}) c = \omega^2 M c. \tag{9}$$

Because E and ρ depend on x and z only, every integral factorises into an x–z part and a y part, and each block of the matrices is a Kronecker product. The x–z integrals are evaluated by Gauss–Legendre quadrature, with additional points for fractional k_x, for which the power law is not a polynomial. The lowest eigenvalues are obtained by shift-invert Lanczos iteration for the 3D model and by a dense solver for the plate models.

To separate flexural from in-plane modes, the fraction of the kinetic energy carried by w is computed for every mode, $\chi = \dfrac{\int \rho w^2 \, dV}{\int \rho (u^2 + v^2 + w^2) \, dV}$. A mode is called flexural if $\chi > 0.5$ and in-plane otherwise. For plates graded only through the thickness the in-plane modes have $\chi = 0$. In bidirectional plates, bending and stretching are coupled along the plate, so the modes are mixed to some degree. For $a/h \geq 10$ the flexural modes computed here have $\chi > 0.8$ and the in-plane modes $\chi < 0.11$. Strongly mixed modes ($0.25 \leq \chi \leq 0.8$) occur only among the higher modes of thick plates ($a/h = 5$), where flexural and in-plane frequencies are close.

4 Numerical Results

The plates are made of alumina (ceramic) and aluminium (metal), with the properties in Table 1 and $\nu = 0.3$. The natural frequencies are reported as

$$\Omega = \omega a^2 \sqrt{\frac{\rho_c h}{D_c}}, \tag{10}$$

so that $\Omega = 2\pi^2 = 19.74$ for the fundamental mode of a thin, simply supported square ceramic plate without a foundation. Unless stated otherwise, the plate is square with $a/h = 10$, $k_x = 2$, $k_z = 1$, $\alpha_p = 0.1$, $(K_u, K_s, K_l) = (100, 10, 100)$ and SSSS edges. The 3D model uses $N = 14$ in-plane terms for SSSS plates and $N = 16$ when an edge is clamped, with $N_z = 10$ through the thickness; the plate models use $N = 16$ ($N = 20$ with clamped edges). In the tables, in-plane modes ($\chi < 0.5$) are marked with an asterisk.

Table 1. Material properties of the constituents

Constituent	E (GPa)	ρ (kg/m^3)	ν
Alumina, Al_2O_3 (ceramic)	380	3,800	0.3
Aluminium, Al (metal)	70	2,707	0.3

Note: E = Young modulus; ρ = mass density; ν = Poisson's ratio.

4.1 Validation and Convergence

Homogeneous plate. For a homogeneous, simply supported square plate with $a/h = 10$, Table 2 compares the frequency parameter $\lambda = \omega h\sqrt{\rho/G}$ with the classical exact 3D values of Srinivas et al. [36]. The independent state-space solution described below and the 3D Ritz model agree with each other to at least nine digits and with the published values to all reported digits (0.09315 rounds to 0.0932 [36]).

Table 2. Homogeneous SSSS plate, $a/h = 10$, $\nu = 0.3$: $\lambda = \omega h\sqrt{\rho/G}$ for modes (m, n)

Solution	(1,1)	(1,2)	(2,2)
Srinivas et al. [36]	0.09320	0.22260	0.34210
Present, exact state-space	0.09315	0.22260	0.34207
Present, 3D Ritz	0.09315	0.22260	0.34207

Note: 3D = three-dimensional; SSSS = all four edges simply supported; a/h = length-to-thickness ratio; ν = Poisson's ratio; λ = dimensionless frequency parameter; ω = circular frequency; ρ = mass density; G = shear modulus; m and n = mode numbers in the x and y directions.

Graded plates on a Kerr foundation. For $k_x = 0$ and hard simply supported edges, the 3D problem has an exact solution. With $u = U(z)\cos(m\pi x/a)\sin(n\pi y/b)$, $v = V(z)\sin(m\pi x/a)\cos(n\pi y/b)$ and $w = W(z)\sin(m\pi x/a)\sin(n\pi y/b)$, the equations of motion reduce to six first-order ordinary differential equations (ODEs) for the state vector $\{U, V, W, \tau_{xz}, \tau_{yz}, \sigma_z\}$. The shear stresses vanish on both faces, $\sigma_z = 0$ on the top face, and the Kerr foundation gives $\sigma_z = k_{\text{eq}} W$ on the bottom face, with $k_{\text{eq}} = k_u(k_l + k_s\lambda_{mn}^2)/(k_u + k_l + k_s\lambda_{mn}^2)$ and $\lambda_{mn}^2 = (m\pi/a)^2 + (n\pi/b)^2$. The system was integrated with an eighth-order Runge–Kutta method at a relative tolerance of 10^{-12}, and the frequencies were found as roots of the 3×3 boundary determinant. This solution shares no code with the Ritz model. Table 3 shows that the 3D Ritz model reproduces it to all reported digits for two thickness ratios, two thickness indices and three foundations.

Convergence. Table 4 lists the first six frequencies of the reference bidirectional plate for increasing N and N_z. For the SSSS plate all six frequencies are converged to the reported digits from $N = 10$. With clamped edges the frequencies decrease monotonically and slowly with N, because the 3D clamped condition prescribes all three displacements over the whole edge face. Between $N = 16$ and $N = 18$ the clamped frequencies still decrease by up to about 0.01, and the successive differences decay approximately as $N^{-2.3}$, which reflects the singular behaviour of the 3D field at the clamped edges. An extrapolation of the sequence $N = 14, 16, 18$ to $N \to \infty$ (rows with $N = \infty$ in Table 4) shows that the clamped values obtained with $N = 16$ exceed the converged values by at most about 0.04. Since Rayleigh–Ritz eigenvalues converge from above, the clamped results are upper bounds that are accurate to about three significant digits, whereas the SSSS results are accurate to all reported digits. The through-thickness expansion $N_z = 10$ is sufficient in all cases: changing N_z from 8 to 12 alters the frequencies by at most one unit in the fourth decimal.

4.2 Accuracy of the Plate Theories

Table 5 compares the first two flexural frequencies from the two plate theories with the 3D solution. The quasi-3D theory overestimates all the frequencies, and the HSDT overestimates the fundamental frequency in every case; for

the second flexural frequency of the thickness-graded SSSS plates ($k_x = 0$), however, the HSDT gives slightly lower values than the 3D solution. The two errors also behave differently as the plate becomes thinner.

Table 3. SSSS, $k_x = 0$, $\alpha_p = 0.1$: present 3D Ritz model versus the exact state-space solution

a/h	k_z	(K_u, K_s, K_l)	Ω_{11} Present	Ω_{11} Exact	Ω_{12} Present	Ω_{12} Exact
5	1	–	13.2795	13.2795	29.5920	29.5920
5	1	(100, 10, 100)	16.1150	16.1150	30.9719	30.9719
5	1	(1000, 10, 100)	20.6036	20.6036	34.9315	34.9315
5	5	–	10.3850	10.3850	22.4119	22.4119
5	5	(100, 10, 100)	14.1586	14.1586	24.4207	24.4207
5	5	(1000, 10, 100)	19.3512	19.3512	29.4386	29.4386
10	1	–	14.3071	14.3071	34.3925	34.3925
10	1	(100, 10, 100)	17.2579	17.2579	35.8308	35.8308
10	1	(1000, 10, 100)	22.1246	22.1246	40.2613	40.2613
10	5	–	11.5117	11.5117	27.2355	27.2355
10	5	(100, 10, 100)	15.4300	15.4300	29.2530	29.2530
10	5	(1000, 10, 100)	21.3234	21.3234	35.1320	35.1320

Note: 3D = three-dimensional; SSSS = all four edges simply supported; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; Ω_{11} and Ω_{12} = the dimensionless frequencies of the (1,1) and (1,2) flexural modes, where $\Omega = \omega a^2(\rho_c h/D_c)^{1/2}$ is the dimensionless frequency; Present = the 3D Rayleigh–Ritz model; Exact = the independent state-space solution; (K_u, K_s, K_l) = the dimensionless Kerr parameters; "–" = a plate without a foundation. All plates are simply supported on all four edges (SSSS), with $k_x = 0$ and $\alpha_p = 0.1$.

Table 4. Convergence of the 3D model; $k_x = 2$, $k_z = 1$, $\alpha_p = 0.1$, $a/h = 10$, Kerr (100, 10, 100)

BC	N	N_z	Ω_1	Ω_2	Ω_3	Ω_4	Ω_5	Ω_6
SSSS	10	10	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3195
SSSS	12	10	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3194
SSSS	14	10	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3194
SSSS	16	10	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3194
SSSS	14	8	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3194
SSSS	14	12	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3194
CSCS	10	10	18.5798	29.7890	35.7717	38.8950*	46.5051	49.2569
CSCS	12	10	18.5718	29.7823	35.7511	38.8950*	46.4871	49.2506
CSCS	14	10	18.5675	29.7787	35.7397	38.8950*	46.4772	49.2471
CSCS	16	10	18.5650	29.7765	35.7326	38.8950*	46.4711	49.2450
CSCS	18	10	18.5633	29.7750	35.7279	38.8950*	46.4670	49.2435
CSCS	∞	10	18.5585	29.7705	35.7129	38.8950*	46.4537	49.2389
CSCS	14	8	18.5675	29.7787	35.7397	38.8950*	46.4773	49.2471
CSCS	14	12	18.5675	29.7787	35.7397	38.8950*	46.4772	49.2471
CCCC	10	10	21.3638	36.5628	37.3551	51.3161	57.6835	60.3496
CCCC	12	10	21.3499	36.5378	37.3304	51.2834	57.6450	60.3095
CCCC	14	10	21.3424	36.5238	37.3168	51.2654	57.6230	60.2877
CCCC	16	10	21.3380	36.5152	37.3084	51.2542	57.6091	60.2741
CCCC	18	10	21.3351	36.5095	37.3029	51.2468	57.6000	60.2651
CCCC	∞	10	21.3265	36.4910	37.2852	51.2227	57.5701	60.2363
CCCC	14	8	21.3424	36.5238	37.3168	51.2654	57.6231	60.2878
CCCC	14	12	21.3424	36.5238	37.3168	51.2654	57.6230	60.2877

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped; 3D = three-dimensional; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; BC = boundary conditions; Ω_1–Ω_6 = the first six dimensionless frequencies; SSSS = all four edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported; CCCC = all four edges clamped; N = the number of in-plane terms in each direction; N_z = the number of through-thickness terms; "*" = marks in-plane modes (kinetic-energy fraction of the transverse displacement below 0.5). The rows with $N = \infty$ give the values extrapolated from the results for $N = 14$, 16 and 18 with $N_z = 10$.

The HSDT error is a pure thickness effect. For SSSS plates it is noticeable at $a/h = 5$, small at $a/h = 10$ and very small at $a/h = 20$. It is somewhat larger for bidirectional plates than for the thickness-graded plate at $a/h = 5$, and clearly larger for CSCS plates, for which it is the largest of all the cases considered at $a/h = 5$. Since the clamped 3D values converge from above (Table 4), these CSCS differences are, if anything, slightly underestimated.

The quasi-3D theory behaves differently. For the thickness-graded plate ($k_x = 0$, $k_z = 1$) its error does not vanish as the plate becomes thin. It remains practically unchanged from $a/h = 10$ to $a/h = 50$, whereas the HSDT error becomes negligible. This persistent error is the signature of thickness locking [37], identified for the same

plates under mechanical loading [30]. The single stretching function gives a transverse normal strain $g'(z)w_z$ that is odd in z. In a graded plate the bending strains are measured from a neutral surface that does not coincide with the mid-plane, so the Poisson contraction through the thickness, $\varepsilon_z \approx -\nu(\varepsilon_x + \varepsilon_y)/(1-\nu)$, has an even part that the theory cannot represent. The in-plane stiffness is then governed by the 3D coefficient C_{11} instead of the plane-stress coefficient Q_{11}, which stiffens the plate. The effect is weaker in bidirectional plates, where the error of thin SSSS plates is small, because for $k_x \geq 2$ most of the plate is nearly metallic and therefore nearly homogeneous through its thickness. For thick plates the quasi-3D error can be smaller than that of the HSDT ($k_x = 5$, $a/h = 5$), but this is a cancellation of two errors of opposite sign rather than a sign of better kinematics.

For free vibration, therefore, both theories are acceptable for engineering purposes when $a/h \geq 10$ and the plate is simply supported. The HSDT is more consistent, because its error vanishes for thin plates. The reduced-unknown quasi-3D theory offers no advantage over the HSDT for these plates, and its error for thickness-graded plates should be kept in mind when it is used to generate reference values. The HSDT is used, together with 3D checks, in the parametric study below.

Table 5. First two flexural frequencies from the plate theories versus 3D elasticity; $\alpha_p = 0.1$, Kerr (100, 10, 100)

BC	a/h	(k_x, k_z)	Ω_1 3D	Ω_1 HSDT	Ω_1 Quasi-3D	Ω_2 3D	Ω_2 HSDT	Ω_2 Quasi-3D
SSSS	5	(0, 1)	16.1150	16.3288	16.4653	30.9719	30.9663	31.5353
SSSS	5	(2, 1)	14.0654	14.3664	14.2542	23.7705	23.9934	24.0208
SSSS	5	(5, 1)	13.4799	13.7791	13.6122	21.5586	21.8105	21.7210
SSSS	5	(2, 5)	13.4376	13.7647	13.6401	21.8094	22.0550	22.0739
SSSS	10	(0, 1)	17.2579	17.3085	17.5235	35.8308	35.7975	36.4700
SSSS	10	(2, 1)	15.0498	15.1208	15.1315	27.7255	27.7648	27.8970
SSSS	10	(5, 1)	14.3860	14.4532	14.4258	25.0514	25.0952	25.1274
SSSS	10	(2, 5)	14.4783	14.5521	14.5603	25.7822	25.8337	25.9592
SSSS	20	(0, 1)	17.5986	17.6111	17.8472	37.6128	37.6009	38.2961
SSSS	20	(2, 1)	15.3409	15.3586	15.4001	29.2278	29.2360	29.3825
SSSS	20	(5, 1)	14.6483	14.6648	14.6721	26.3863	26.3954	26.4462
SSSS	20	(2, 5)	14.7826	14.8006	14.8429	27.3259	27.3366	27.4896
CSCS	5	(0, 1)	19.7347	20.4575	20.8801	32.4963	33.1608	33.8961
CSCS	5	(2, 1)	16.3924	17.0910	17.1970	24.7690	25.5533	25.7393
CSCS	5	(5, 1)	15.6193	16.1757	16.1930	22.4517	23.0720	23.1070
CSCS	5	(2, 5)	15.3242	15.9787	16.0403	22.7019	23.3985	23.5097
CSCS	10	(0, 1)	22.4465	22.6937	23.1578	38.6817	39.0141	39.8489
CSCS	10	(2, 1)	18.5650	18.8143	18.9759	29.7765	30.1525	30.4080
CSCS	10	(5, 1)	17.5729	17.7554	17.8488	26.8278	27.1018	27.2259
CSCS	10	(2, 5)	17.4194	17.6556	17.8012	27.5091	27.8509	28.0781
CSCS	20	(0, 1)	23.4183	23.4645	23.9073	41.1945	41.2895	42.1181
CSCS	20	(2, 1)	19.3789	19.4354	19.5839	31.9096	32.0244	32.2561
CSCS	20	(5, 1)	18.3010	18.3403	18.4299	28.6934	28.7756	28.8855
CSCS	20	(2, 5)	18.2405	18.2967	18.4433	29.6582	29.7691	29.9964

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped; 3D = three-dimensional; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; BC = boundary conditions; Ω_1 and Ω_2 = the first two flexural dimensionless frequencies; HSDT = higher-order shear deformation; Quasi-3D = reduced-unknown quasi-3D theory theory; SSSS = all four edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported.

4.3 Benchmark Results

Tables 6 and 7 give new 3D results for future comparison: the first six frequencies, with in-plane modes marked, for plates on the reference Kerr foundation (SSSS, CSCS and CCCC edges) and for SSSS plates without a foundation. Two porosities, two thickness indices and four length indices are covered. Based on Table 4, the SSSS values are accurate to all reported digits, and the values for plates with clamped edges to about three significant digits (as Rayleigh–Ritz values, they are upper bounds of the converged frequencies).

Several features of these results are worth noting. On the foundation, the in-plane modes of SSSS plates appear as the fourth and fifth modes, and those of CSCS plates as the fourth mode; they are almost unaffected by the foundation (compare Tables 6 and 7) and fall rapidly with k_x. In CCCC plates the in-plane modes lie above the first six flexural modes. For $k_x = 0$ the second and third modes of SSSS and CCCC plates are degenerate; for $k_x > 0$ they separate, and the separation grows with k_x. Porosity lowers the frequencies of the unsupported plates in all cases, whereas on the foundation its effect on the fundamental frequency of the thickness-graded SSSS plate is negligible (17.2541 vs. 17.2579).

4.4 Parametric Study

Gradation indices. Figure 2 shows the effect of the two indices on the fundamental frequency (HSDT lines, 3D symbols) and on the location $x_{\max}$ of the peak of the first mode along $y = b/2$. The frequency decreases

monotonically with both indices. In-plane grading replaces ceramic by aluminium, whose stiffness-to-density ratio E/ρ is about four times lower than that of alumina, so the plate becomes softer faster than it becomes lighter. For $k_z = 1$ the 3D fundamental frequency falls from 17.26 at $k_x = 0$ to 15.05 at $k_x = 2$ and 14.39 at $k_x = 5$ on the foundation, a moderate reduction. Without a foundation the reduction is much larger, from 14.31 to 9.64, i.e. about one third, because the foundation stiffness does not depend on the composition. The steepest drop occurs for $k_x \leq 1$; beyond $k_x \approx 2$ the plate is mostly metallic and the frequency changes slowly.

Table 6. 3D benchmark frequencies; $a/h = 10$, Kerr (100, 10, 100)

BC	α_p	k_z	k_x	Ω_1	Ω_2	Ω_3	Ω_4	Ω_5	Ω_6
SSSS	0.0	1	0	17.2541	36.3790	36.3790	53.5156*	53.5156*	55.0008
SSSS	0.0	1	1	15.8434	31.6590	31.7062	43.6125*	44.4200*	47.3622
SSSS	0.0	1	2	15.2405	29.3863	29.5845	40.6709*	41.3203*	44.0543
SSSS	0.0	1	5	14.6266	27.0725	27.4772	36.5514*	38.3674*	40.5986
SSSS	0.0	5	0	15.7881	31.1649	31.1649	41.7303*	41.7303*	46.1020
SSSS	0.0	5	1	15.0994	28.9306	28.9821	37.3043*	37.4859*	42.6003
SSSS	0.0	5	2	14.7470	27.7491	27.8233	35.9262*	36.3293*	40.9178
SSSS	0.0	5	5	14.3715	26.4286	26.6473	34.3078*	35.1538*	39.0876
SSSS	0.1	1	0	17.2579	35.8308	35.8308	53.5088*	53.5088*	54.0641
SSSS	0.1	1	1	15.7143	30.4088	30.4588	41.7582*	42.9276*	45.2425
SSSS	0.1	1	2	15.0498	27.7255	27.9721	38.4081*	38.8950*	41.3194
SSSS	0.1	1	5	14.3860	25.0514	25.5310	33.3552*	35.2034*	37.2307
SSSS	0.1	5	0	15.4300	29.2530	29.2530	39.9613*	39.9613*	42.8949
SSSS	0.1	5	1	14.8238	27.0385	27.1091	34.4487*	34.7441*	39.3316
SSSS	0.1	5	2	14.4783	25.7822	25.8427	32.7793*	33.1878*	37.4846
SSSS	0.1	5	5	14.1020	24.3209	24.5566	30.7274*	31.6721*	35.4338
CSCS	0.0	1	0	22.6152	39.2733	47.3508	53.5156*	62.5168	68.4222
CSCS	0.0	1	1	20.0292	34.0521	40.4298	43.6125*	53.3090	58.3739
CSCS	0.0	1	2	19.2095	31.6105	38.0470	41.3203*	49.6855	53.1440
CSCS	0.0	1	5	18.3353	29.0797	35.5537	38.3674*	45.8613	48.1954
CSCS	0.0	5	0	19.8701	33.3264	39.4080	41.7303*	51.5532	56.6418
CSCS	0.0	5	1	18.6724	30.9726	36.4207	37.3043*	47.6354	52.3970
CSCS	0.0	5	2	18.2212	29.6961	35.2362	36.3293*	45.8708	49.8450
CSCS	0.0	5	5	17.7219	28.2744	33.9063	35.1538*	43.8898	47.1330
CSCS	0.1	1	0	22.4465	38.6817	46.6429	53.5088*	61.5218	67.2651
CSCS	0.1	1	1	19.5096	32.6522	38.5730	41.7582*	50.7891	55.4963
CSCS	0.1	1	2	18.5650	29.7765	35.7326	38.8950*	46.4711	49.2450
CSCS	0.1	1	5	17.5729	26.8278	32.7906	35.2034*	41.9238	43.4749
CSCS	0.1	5	0	19.0452	31.2291	36.8058	39.9613*	47.9353	52.5791
CSCS	0.1	5	1	17.8831	28.8803	33.6948	34.4487*	43.8465	48.2218
CSCS	0.1	5	2	17.4194	27.5091	32.4045	33.1878*	41.9061	45.3921
CSCS	0.1	5	5	16.8926	25.9209	30.9100	31.6721*	39.6569	42.2959
CCCC	0.0	1	0	26.9220	49.6013	49.6013	69.2348	81.6650	82.4035
CCCC	0.0	1	1	23.6411	42.3454	42.6280	59.0121	68.9714	69.4776
CCCC	0.0	1	2	22.3022	39.0488	39.7465	54.7868	62.4589	64.6548
CCCC	0.0	1	5	20.8932	35.5030	37.0162	50.2457	56.5494	59.9044
CCCC	0.0	5	0	23.2387	41.1465	41.1465	56.5383	66.0695	66.7244
CCCC	0.0	5	1	21.7421	38.0297	38.1921	52.2836	61.0631	61.6386
CCCC	0.0	5	2	21.0438	36.4884	36.7607	50.3269	58.3569	59.2582
CCCC	0.0	5	5	20.2676	34.6341	35.3354	48.1178	55.3775	56.8868
CCCC	0.1	1	0	26.6373	48.8590	48.8590	68.1777	80.4481	81.1669
CCCC	0.1	1	1	22.8924	40.4052	40.7657	56.2899	65.5238	65.9564
CCCC	0.1	1	2	21.3380	36.5152	37.3084	51.2542	57.6091	60.2741
CCCC	0.1	1	5	19.7169	32.3681	34.0803	45.8241	50.7494	54.6162
CCCC	0.1	5	0	22.0711	38.4035	38.4035	52.5487	61.3362	61.9445
CCCC	0.1	5	1	20.5961	35.1490	35.3718	48.0627	55.9862	56.5228
CCCC	0.1	5	2	19.8603	33.4969	33.7614	45.8926	52.9199	53.8563
CCCC	0.1	5	5	19.0136	31.3879	32.1552	43.3653	49.4759	51.1602

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped; 3D = three-dimensional; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; BC = boundary conditions; Ω_1–Ω_6 = the first six dimensionless frequencies; SSSS = all four edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported; CCCC = all four edges clamped; "*" = marks in-plane modes. Values for plates with clamped edges are upper bounds accurate to about three significant digits.

In-plane grading also makes the modes asymmetric. The peak of the first mode moves toward the metal-rich edge $x = 0$, reaching $x_{\max}/a = 0.41$–0.46 at $k_x \approx 2$–2.5 depending on k_z, and then moves back toward the centre as k_x increases further (Figure 2c). A mode concentrates where the local ratio of stiffness to mass, and hence the local frequency, is lowest; this is the metal-rich part. For large k_x only a narrow strip near $x = a$ is ceramic-rich, the plate is nearly uniform again, and the asymmetry decreases. The shift is largest for small k_z, where the top layer is most ceramic-rich and the contrast along the plate is strongest. The foundation reduces the shift slightly (for $k_z = 1$, $x_{\max}/a = 0.435$ on the foundation versus 0.428 without it at $k_x = 2$), because its stiffness raises the local frequency more where the plate is lighter. The 3D results (symbols) confirm the HSDT trends.

Mode shapes. Figure 3 shows the first four flexural modes from the 3D model for $k_x = 0, 2$ and 5. For the thickness-graded plate the second and third modes are degenerate (35.831), as for any square plate with symmetric properties. In-plane grading breaks the symmetry between x and y and splits the pair: for $k_x = 2$, the mode with a nodal line parallel to the x-axis has $\Omega = 27.725$, and the mode with a nodal line parallel to the y-axis has

$\Omega = 27.972$. The nodal lines parallel to the y-axis move toward the ceramic-rich edge (to $x/a \approx 0.45$ in the third mode for $k_x = 2$), and the larger antinode lies in the metal-rich half. The same pattern appears in the fourth mode. A plate theory solved by a double Fourier series assumes symmetric mode shapes and cannot represent this distortion.

Table 7. 3D benchmark frequencies; $a/h = 10$, SSSS, no foundation

BC	α_p	k_z	k_x	Ω_1	Ω_2	Ω_3	Ω_4	Ω_5	Ω_6
SSSS	0.0	1	0	14.6196	35.1022	35.1022	53.5156*	53.5156*	54.1593
SSSS	0.0	1	1	12.5884	30.0164	30.0570	43.6125*	44.4196*	46.2690
SSSS	0.0	1	2	11.6426	27.5054	27.7460	40.6707*	41.3203*	42.8249
SSSS	0.0	1	5	10.6503	24.9349	25.4084	36.5498*	38.3674*	39.1999
SSSS	0.0	5	0	12.4543	29.4795	29.4795	41.7303*	41.7303*	44.9704
SSSS	0.0	5	1	11.4020	27.0139	27.0651	37.3043*	37.4850*	41.3039
SSSS	0.0	5	2	10.8521	25.6944	25.7873	35.9252*	36.3293*	39.5386
SSSS	0.0	5	5	10.2588	24.2207	24.4775	34.3034*	35.1538*	37.6111
SSSS	0.1	1	0	14.3071	34.3925	34.3925	53.1180	53.5088*	53.5088*
SSSS	0.1	1	1	11.9591	28.4807	28.5191	41.7582*	42.9269*	43.9600
SSSS	0.1	1	2	10.8262	25.4485	25.7617	38.4077*	38.8950*	39.8397
SSSS	0.1	1	5	9.6393	22.3907	22.9772	33.3531*	35.2034*	35.4932
SSSS	0.1	5	0	11.5117	27.2355	27.2355	39.9613*	39.9613*	41.5402
SSSS	0.1	5	1	10.4511	24.7095	24.7802	34.4487*	34.7421*	37.7498
SSSS	0.1	5	2	9.8399	23.2513	23.3387	32.7775*	33.1878*	35.7789
SSSS	0.1	5	5	9.1634	21.5561	21.8533	30.7184*	31.6721*	33.5775

Note: 3D = three-dimensional; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; BC = boundary conditions; Ω_1–Ω_6 = the first six dimensionless frequencies; SSSS = all four edges simply supported; "*" = marks in-plane modes.

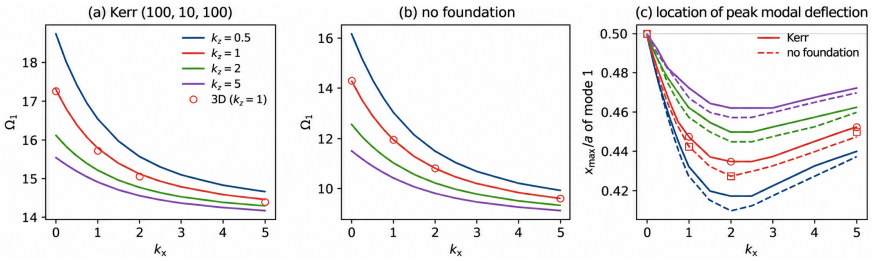


Figure 2. Effect of the gradation indices on: (a) the fundamental frequency on the Kerr foundation; (b) the fundamental frequency without a foundation; (c) the location of the peak of the first mode along $y = b/2$; SSSS, $a/h = 10$, $\alpha_p = 0.1$, Lines: HSDT, symbols: 3D

Note: SSSS = all four edges simply supported; a/h = length-to-thickness ratio; α_p = porosity volume fraction; k_x = gradation index along the length; HSDT = higher-order shear deformation theory; 3D = three-dimensional.

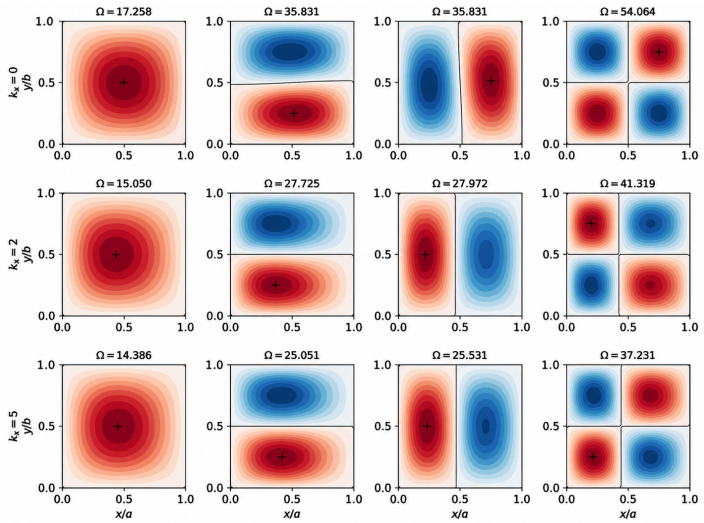


Figure 3. First four flexural mode shapes (mid-plane deflection, 3D model) of plates on the Kerr foundation (100, 10, 100) for $k_x = 0$ (top), 2 (middle) and 5 (bottom); SSSS, $a/h = 10$, $k_z = 1$, $\alpha_p = 0.1$

Note: SSSS = all four edges simply supported; a/h = length-to-thickness ratio; α_p = porosity volume fraction; k_x = gradation index along the length; k_z = gradation index through the thickness; 3D = three-dimensional; black lines = nodal lines; crosses = peak amplitude.

Kerr foundation. Figure 4 shows the effect of the three Kerr parameters on the fundamental flexural frequency and on the location of its peak. As K_u increases, the frequency rises from its value without a foundation (10.82) to the Pasternak limit with the same K_s and K_l (dotted lines). For $K_s = 0$ it rises to the Winkler limit with modulus K_l (16.30). The upper spring thus acts in series with the two lower layers, as in static problems [30]. For $K_u \lesssim 30$ the upper spring governs and the other two parameters hardly matter. For $K_u \gtrsim 300$ the shear layer and the lower spring control the response and can raise the frequency by a factor of up to five. The 3D results (stars) confirm the HSDT trend: 11.46, 15.05, 21.25 and 23.20 at K_u = 10, 10^2, 10^3, and 10^4, versus 11.47, 15.12, 21.43, and 23.41 from the HSDT.

A stiff foundation has two further effects that are specific to bidirectional plates. First, it moves the peak of the first mode from the metal-rich side across the centre. For $(K_s, K_l) = (100, 100)$ the peak reaches $x_{\max}/a = 0.51$ at large K_u, on the ceramic-rich side. When the foundation dominates the stiffness, the local frequency scales with $\sqrt{k_{\text{eq}}/(\rho h)}$ and is lowest where the plate is heaviest, i.e. near the ceramic-rich edge. Second, the foundation raises only the flexural frequencies. The first in-plane mode ($\Omega \approx 38.4$, chain line) is practically unaffected, so for stiff foundations it becomes the fundamental mode of the plate. This occurs for K_u between about 1.6×10^3 and 2.5×10^3 when $(K_s, K_l) = (100, 100)$, and between 2.5×10^3 and 4×10^3 when $(K_s, K_l) = (10, 1000)$. Near the crossing the in-plane frequency changes slightly (between 38.1 and 38.6), because in-plane grading couples bending and stretching. Analyses that track only the transverse displacement, or that assume the lowest mode to be flexural, miss this change of the fundamental mode.

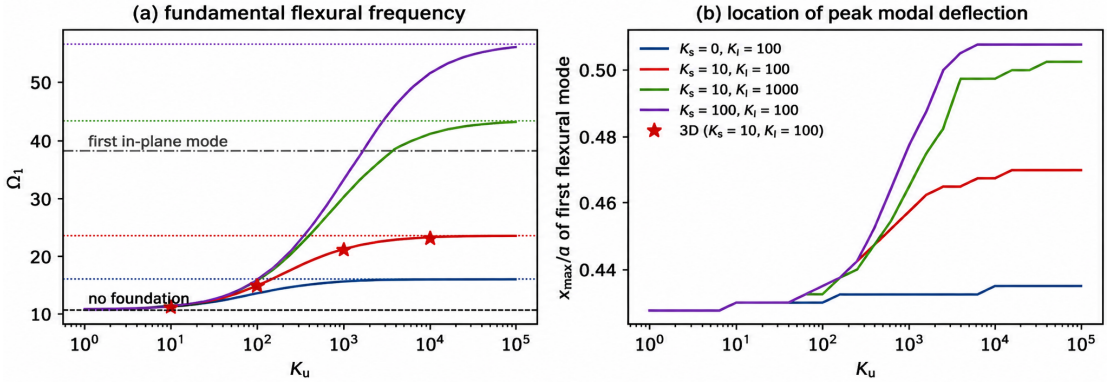


Figure 4. Effect of the Kerr parameters on: (a) the fundamental flexural frequency; (b) the location of its peak along $y = b/2$; HSDT, SSSS, $a/h = 10$, $k_x = 2$, $k_z = 1$, $\alpha_p = 0.1$

Note: SSSS = all four edges simply supported; HSDT = higher-order shear deformation theory; a/h = length-to-thickness ratio; α_p = porosity volume fraction; k_x = gradation index along the length; k_z = gradation index through the thickness; dotted lines = Pasternak/Winkler limits ($K_u \to \infty$); dashed line = no foundation; chain line = first in-plane mode; stars = three-dimensional (3D) results.

Porosity. Porosity lowers both the modulus and the density, and Table 8 and Figure 5 show that its net effect depends on the foundation. Without a foundation the frequency decreases with α_p, because Eq. (2) removes a larger fraction of the stiffness than of the mass, particularly in the metal-rich regions. Raising α_p from 0 to 0.2 lowers the 3D frequency only slightly for $k_x = 0$, but considerably for $k_x = 2$ and even more for $k_x = 5$, since bidirectional plates are mostly metallic. On the reference Kerr foundation, part of the stiffness is supplied by the foundation and is not affected by porosity. The decrease is then small for the bidirectional plates, and the frequency of the thickness-graded plate is practically independent of α_p (17.25–17.26 for $\alpha_p \leq 0.15$). On the stiff foundation (1000, 100, 1000) the trend reverses. The foundation dominates the stiffness, porosity mainly reduces the mass, and the fundamental flexural frequency increases appreciably with α_p for both $k_x = 0$ and $k_x = 2$. The in-plane frequencies, which the foundation does not raise, decrease with porosity. As a result, for $k_x = 5$ with $\alpha_p \geq 0.1$, and for $k_x = 2$ with $\alpha_p = 0.2$, an in-plane mode becomes the fundamental mode (marked † in Table 8). Porosity therefore cannot be said to lower the natural frequencies in general: the sign of its effect is set by the ratio of foundation to plate stiffness. The HSDT reproduces all these trends; its differences from the 3D results are negligible without a foundation, small on the reference foundation and somewhat larger on the stiff foundation.

Edge conditions. Table 9 lists HSDT results for seven edge combinations. As expected, clamping raises the frequencies, and the CCCC plate has the highest fundamental frequency (26.77, 21.52 and 19.86 for $k_x = 0, 2$ and 5). For the thickness-graded plate the CSCS and SCSC plates are identical by symmetry. In a bidirectional plate they differ: clamping the graded edges $x = 0, a$ (CSCS) gives a higher fundamental frequency (18.81 for $k_x = 2$) than clamping the ungraded edges $y = 0, b$ (SCSC, 18.65), but a lower second frequency. SCSC also gives the largest shift of the peak toward the metal-rich edge ($x_{\max}/a = 0.408$ for $k_x = 2$), because the simply supported graded edges leave the soft metal-rich region free to rotate. The CFCF plate has closely spaced first and second frequencies

(15.64 and 17.87 for $k_x = 2$). In all cases the peak of the first mode lies in the metal-rich half of the plate.

Table 8. Effect of porosity on the fundamental flexural frequency; SSSS, $a/h = 10$, $k_z = 1$

k_x	α_p	No fdn: 3D	HSDT	(100, 10, 100): 3D	HSDT	(1000, 100, 1000): 3D	HSDT
0	0	14.6196	14.5984	17.2541	17.2950	32.3795	32.7578
0	0.05	14.4766	14.4549	17.2596	17.3050	32.9685	33.3767
0	0.1	14.3071	14.2848	17.2579	17.3085	33.6029	34.0469
0	0.15	14.1037	14.0807	17.2462	17.3029	34.2871	34.7755
0	0.2	13.8567	13.8326	17.2206	17.2846	35.0245	35.5708
2	0	11.6426	11.6362	15.2405	15.2950	33.0444	33.4304
2	0.05	11.2733	11.2688	15.1560	15.2178	33.7582	34.1832
2	0.1	10.8262	10.8242	15.0498	15.1208	34.5321	35.0029
2	0.15	10.2688	10.2703	14.9124	14.9952	35.3674	35.8472
2	0.2	9.5421	9.5489	14.7256	14.8252	36.2631†	37.0209†
5	0	10.6503	10.6445	14.6266	14.6791	33.2709	33.6410
5	0.05	10.1937	10.1894	14.5174	14.5765	33.9956	34.3190
5	0.1	9.6393	9.6368	14.3860	14.4532	34.8975†	35.4499†
5	0.15	8.9453	8.9449	14.2235	14.3010	35.7568†	36.4021†
5	0.2	8.0343	8.0365	14.0141	14.1060	36.6638†	37.4957†

Note: SSSS = all four edges simply supported; HSDT = higher-order shear deformation theory; a/h = length-to-thickness ratio; α_p = porosity volume fraction; k_x = gradation index along the length; k_z = gradation index through the thickness; 3D = three-dimensional; No fdn = a plate without a foundation; (100, 10, 100) and (1000, 100, 1000) = the Kerr parameters (K_u, K_s, K_l); † = cases in which an in-plane mode has a lower frequency than the first flexural mode.

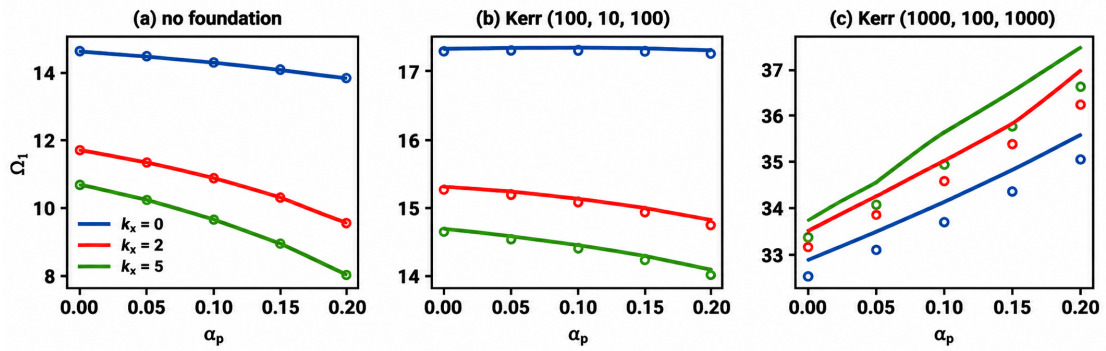


Figure 5. Effect of porosity on the fundamental flexural frequency: (a) without a foundation; (b) on the Kerr foundation (100, 10, 100); (c) on the stiff Kerr foundation (1000, 100, 1000); SSSS, $a/h = 10$, $k_z = 1$

Note: SSSS = all four edges simply supported; a/h = length-to-thickness ratio; α_p = porosity volume fraction; k_x = gradation index along the length; k_z = gradation index through the thickness; (100, 10, 100) and (1000, 100, 1000) = the Kerr parameters (K_u, K_s, K_l).

Table 9. Effect of the edge conditions; HSDT, $a/h = 10$, $k_z = 1$, $\alpha_p = 0.1$, Kerr (100, 10, 100)

BC	$k_x = 0$			$k_x = 2$			$k_x = 5$		
	Ω_1	Ω_2	$x_{\max}/a$	Ω_1	Ω_2	$x_{\max}/a$	Ω_1	Ω_2	$x_{\max}/a$
SSSS	17.3085	35.7975	0.500	15.1208	27.7648	0.435	14.4532	25.0952	0.453
SSSF	12.7315	22.2917	0.500	12.0462	18.6248	0.455	11.8534	17.5149	0.468
CSCS	22.6937	39.0141	0.500	18.8143	30.1525	0.440	17.7554	27.1018	0.453
SCSC	22.6937	39.0141	0.500	18.6516	30.3677	0.408	17.2718	27.6303	0.435
CSCF	19.0587	27.0377	0.500	16.2823	21.9272	0.445	15.6677	20.4027	0.455
CFCF	18.1821	21.1933	0.500	15.6370	17.8654	0.448	15.1094	17.0557	0.455
CCCC	26.7696	49.5027	0.500	21.5226	37.1807	0.430	19.8571	32.9194	0.448

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped, and F = free; HSDT = higher-order shear deformation theory; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; $x_{\max}$ = location of the peak of the first mode along $y = b/2$; BC = boundary conditions; Ω_1 and Ω_2 = the first two flexural dimensionless frequencies; SSSS = all four edges simply supported; SSSF = edge $y = b$ free, other edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported; SCSC = edges $y = 0$ and $y = b$ clamped, edges $x = 0$ and $x = a$ simply supported; CSCF = edges $x = 0$ and $x = a$ clamped, edge $y = 0$ simply supported, edge $y = b$ free; CFCF = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ free; CCCC = all edges clamped.

Thickness ratio. Table 10 compares the three models over a wide range of thickness ratios. The HSDT converges to the 3D solution as the plate becomes thin, and the two practically coincide at $a/h = 50$ for both $k_x = 0$ and

$k_x = 5$. The quasi-3D theory converges to a value that remains noticeably too high for $k_x = 0$. For thick plates ($a/h = 4$) both theories clearly overestimate the frequency.

Table 10. Fundamental frequency versus thickness ratio; SSSS, $k_z = 1$, $\alpha_p = 0.1$, Kerr (100, 10, 100)

k_x	a/h	3D	HSDT	Quasi-3D
0	4	15.4143	15.7630	15.8446
0	5	16.1150	16.3288	16.4653
0	10	17.2579	17.3085	17.5235
0	20	17.5986	17.6111	17.8472
0	50	17.6992	17.7012	17.9433
5	4	12.8248	13.0941	12.9513
5	5	13.4799	13.7791	13.6122
5	10	14.3860	14.4532	14.4258
5	20	14.6483	14.6648	14.6721
5	50	14.7259	14.7285	14.7457

Note: 3D = three-dimensional; SSSS = all four edges simply supported; a/h = length-to-thickness ratio; k_x = gradation index along the length; k_z = gradation index through the thickness; α_p = porosity volume fraction; HSDT = higher-order shear deformation; Quasi-3D = reduced-unknown quasi-3D theory theory.

5 Conclusions

The free vibration of length- and thickness-graded porous plates resting on a three-parameter Kerr foundation was studied with a 3D elasticity Ritz model and two plate theories. The main conclusions are:

1. A 3D Ritz model with a statically condensed Kerr foundation reproduces an independent exact state-space solution to all reported digits and the classical results of Srinivas et al. for homogeneous plates. New 3D benchmark frequencies are provided for bidirectional porous plates with and without a Kerr foundation (Tables 6 and 7).
2. In-plane grading lowers the natural frequencies, because aluminium has a much lower stiffness-to-density ratio than alumina. From $k_x = 0$ to $k_x = 5$ the fundamental frequency falls by about one third without a foundation, and moderately on the reference Kerr foundation.
3. In-plane grading makes the mode shapes asymmetric. The peak of the first mode moves toward the metal-rich edge (to $x/a \approx 0.41$–0.46 for $k_x \approx 2$), degenerate mode pairs of square plates split, and nodal lines move toward the ceramic-rich edge. A stiff foundation moves the peak back across the centre toward the heavier, ceramic-rich edge.
4. The HSDT agrees closely with the 3D solution for SSSS plates with $a/h \geq 10$, and its error vanishes for thin plates. The reduced-unknown quasi-3D theory is not more accurate. For thickness-graded plates it retains a noticeable error even for $a/h = 50$, a consequence of thickness locking. This error decreases with in-plane grading.
5. The Kerr upper spring acts in series with the shear layer and the lower spring. Stiff foundations raise only the flexural frequencies, so the first in-plane mode, which is coupled to bending in bidirectional plates, can become the fundamental mode of the plate.
6. The effect of porosity depends on the foundation. It lowers the frequency of an unsupported plate, considerably so for strongly graded bidirectional plates, has little effect on the reference foundation and raises the flexural frequency on a stiff foundation.
7. In bidirectional plates, clamping the graded edges and clamping the ungraded edges give different frequencies and mode shapes, while the two cases coincide for thickness-graded plates.

For sensitive structural designs involving thick plates, clamped edges or stiff foundations, simplified 2D and reduced-unknown quasi-3D models should be checked against exact 3D elasticity or benchmarked Ritz solutions, since thickness locking and the coupling of bending and stretching can affect the predicted frequencies and modes.

Further work can address forced and transient vibration, thermal pre-stress, and the vibration of plates on tensionless foundations, for which the contact region must be determined as part of the solution.

Data Availability

The Python code used to generate all results and figures is available from the author on reasonable request.

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Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI tools (Gemini) were used exclusively for grammatical checking, stylistic refinement, and linguistic phrasing.

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